

5-Mark Questions & Answers — Data Structures Module 3: Linked Lists

Q1. Write complete C functions for Singly Linked List: Creation, Insertion, Deletion, and Traversal.

Complete working C program with `struct Node` and pointer manipulation functions.

Q2. Explain Queue implementation using a Singly Linked List with Front and Rear pointers in C.

- `enqueue()` inserts at `rear->next` in **O(1)** time.
 - `dequeue()` removes from `front` in **O(1)** time.
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Q3. Explain Polynomial Addition using Linked Lists with an algorithm and diagram.

Traverse both polynomial lists simultaneously:

- If `exp1 == exp2` : Add coefficients, insert term into result list, advance both pointers.
 - If `exp1 > exp2` : Insert term 1 into result list, advance pointer 1.
 - If `exp2 > exp1` : Insert term 2 into result list, advance pointer 2.
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